

Volterra Battery Systems, Inc. - Cell Qualification Test Report

Document number: QR-2170-B-001

Document status: Approved

Product: VX-2170 Revision B

Product name: VX-2170 Cylindrical NMC Cell

Test period: 2026-01-19 to 2026-01-30

Test site: Volterra Holland Cell Plant - Cell Test Laboratory

Test procedures: IEC 61960-3 (rated capacity), Volterra TP-CELL-004 (dimensions), TP-CELL-007 (rate capability)

Summary

Item	Result
Rated capacity (declared)	5.0 Ah
Nominal voltage	3.6 V
Rated energy	18.00 Wh
Result against internal specification	Meets specification

This report applies only to VX-2170 Revision B as built on the qualification lot.

1. Cell identification and dimensions

Form factor: 21700 cylindrical

Chemistry: NMC811 cathode / graphite anode

Nominal voltage: 3.6 V

Charge voltage: 4.2 V

Discharge cut-off voltage: 2.5 V

Measured external dimensions: Diameter 21.0 mm, height 70.0 mm

External volume: 24.25 mL

Mass (mean of samples): 68.5 g

External volume is calculated from measured external dimensions excluding terminals.

2. Capacity and energy

Procedure: Charge CC-CV to charge voltage at 25 C; rest 1 h; discharge at 0.2C to cut-off voltage

Sample	Discharge capacity (Ah)
S01	5.04
S02	5.07
S03	5.05
S04	5.08
S05	5.06
S06	5.03
S07	5.09
S08	5.06
S09	5.05
S10	5.07

Mean measured discharge capacity: 5.06 Ah

Declared rated capacity: 5.0 Ah

Rated energy (nominal voltage x rated capacity): 18.00 Wh

Volumetric energy density (rated energy / external volume): 742 Wh/L

3. Rate capability

Maximum continuous discharge current: 20 A

Maximum continuous discharge power at nominal voltage: 72 W

Capacity-to-power ratio (rated energy / max continuous power): 0.25 : 1

4. Sign-off

Test engineer: Priya Natarajan, Cell Test Engineer (signed 2026-01-30)

Reviewer: Daniel Okafor, Quality Manager (signed 2026-02-03)

Approval date: 2026-02-03